DIGITAL IMAGE PROCESSING

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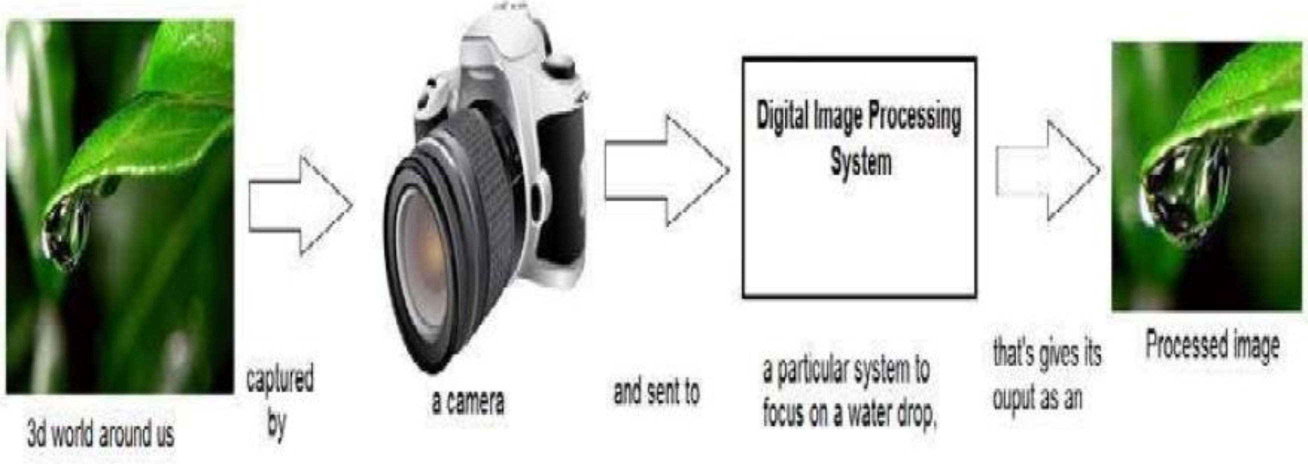
Digital Image Processing means processing digital image by means of a digital computer.

It covers low-, mid-, and high-level processes

- ✓ **low-level:** inputs and outputs are images
- ✓ **mid-level:** outputs are attributes extracted from input images
- ✓ **high-level:** an ensemble of recognition of individual objects


3d world around us
captured by
a camera
and sent to
Digital Image Processing System
a particular system to focus on a water drop,
that's gives its ouput as an
Processed image

Why we need image processing

- ✓ Improvement of pictorial information for human perception.
- ✓ Image processing for autonomous machine application
- ✓ Efficient storage and transmission

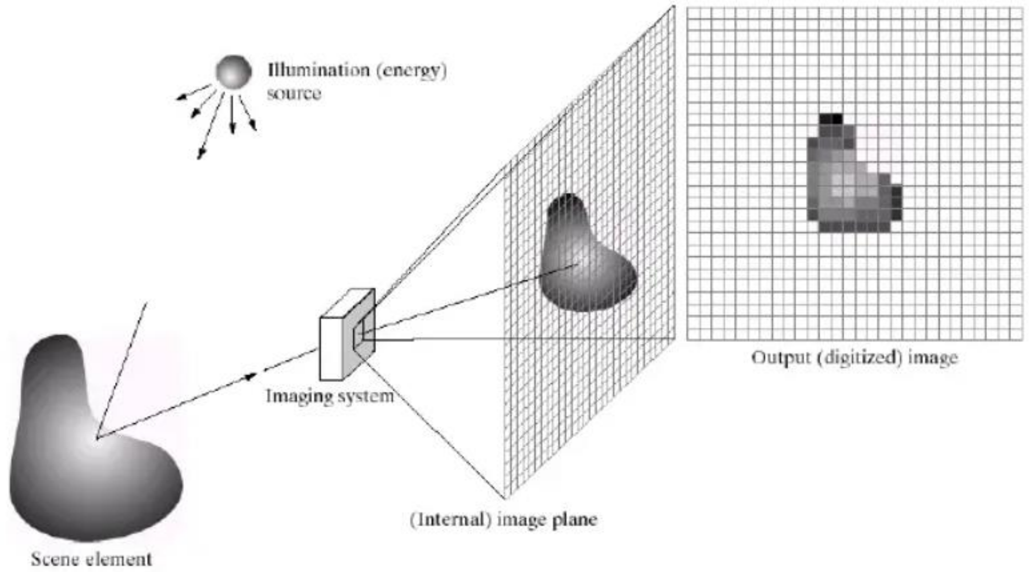
Image:-

- ✓ Projection of 3D scene into a 2D plane.
- ✓ 2D function f(x,y) where x,y are spatial coordinates, and f is at any pair of coordinates (x,y) is called intensity or gray level of image at that particular point.
- ✓ Two types

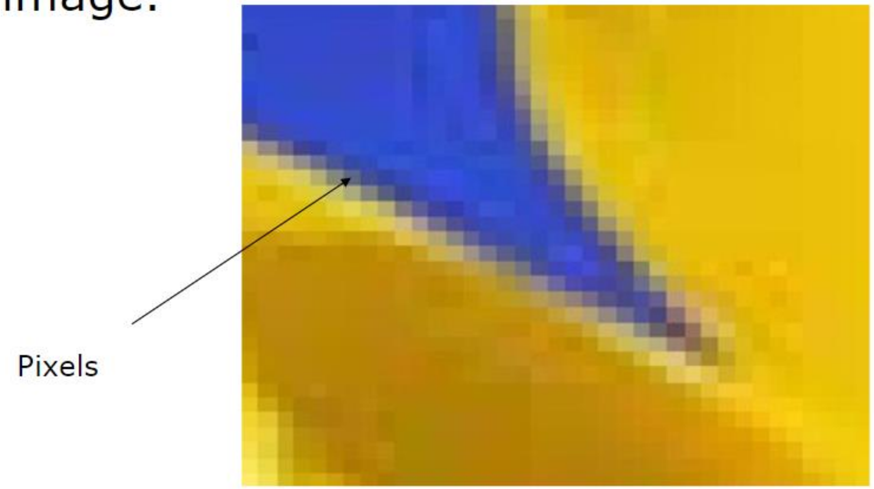
  1. **Analog Image**: f, x,y has continuous range of values
  2. **Digital Image:** when f,x,y are finite.

What is a digital image?

A **digital image** is a representation of a two-dimensional image as a finite set of digital values, called picture elements or pixels



- Pixel: The elements of a digital image.



- A digital image is a representation of a real image as a set of numbers that can be stored and handled by a digital computer.
- In order to translate the image into numbers, it is divided into small areas called **pixels** (picture elements).
- A pixel is the smallest unit of a digital image or graphic that can be displayed and represented on a digital display device.
- The numbers are arranged in an array of rows and columns that correspond to the vertical and horizontal positions of the pixels in the image.

Types of Image

- **BINARY IMAGE**– The binary image contains only two pixel elements, 0 refers to black and 1 refers to white. This image is also known as Monochrome.

- **BLACK AND WHITE IMAGE**– The image which consist of only black and white color is called BLACK AND WHITE IMAGE.

- **8 bit COLOR FORMAT**– It is the most famous image format. It has 256 different shades of colors in it and commonly known as Grayscale Image. In this format, 0 stands for Black, and 255 stands for white, and 127 stands for gray.

- **16 bit COLOR FORMAT**– It is a color image format. It has 65,536 different colors in it. It is also known as High Color Format.

Image Acquisition Process

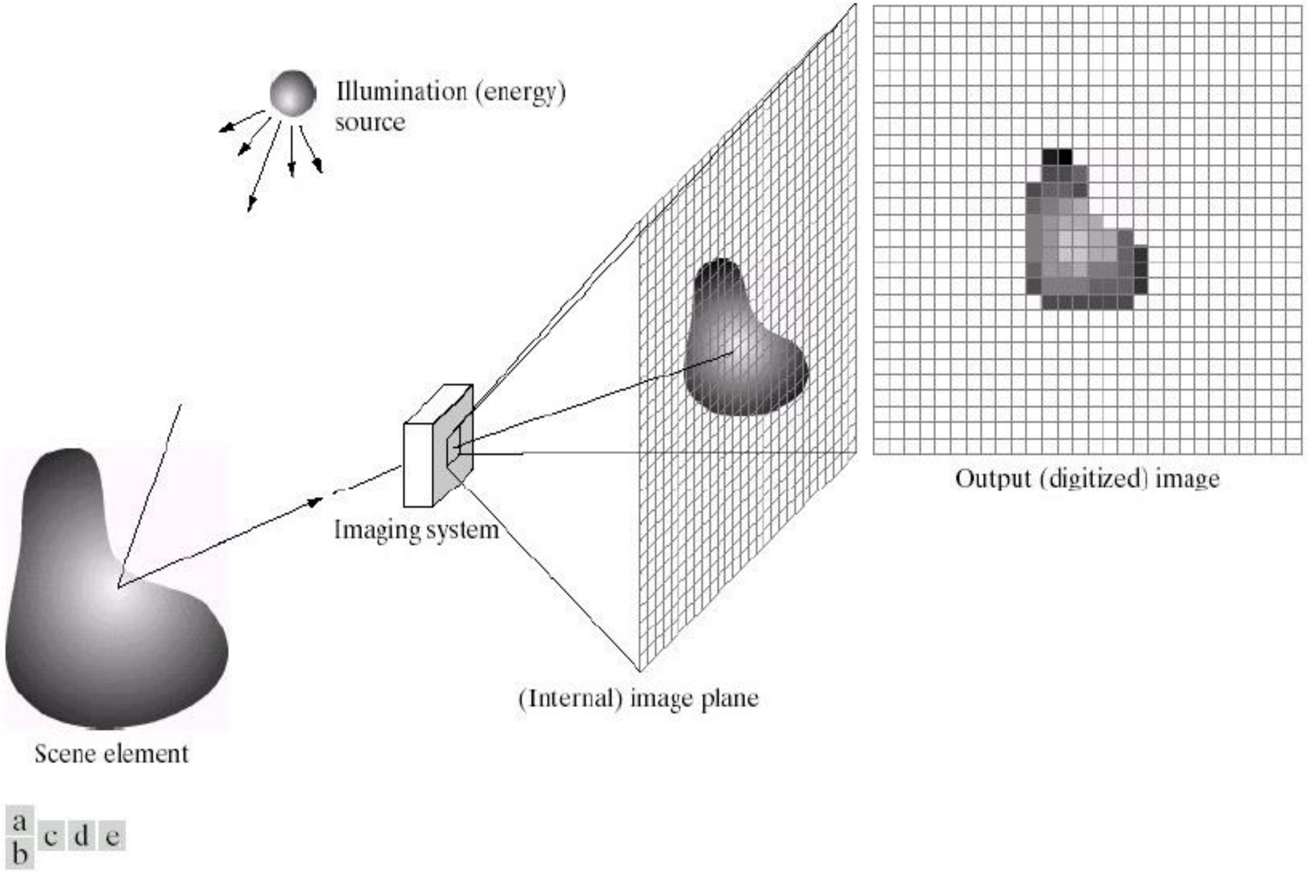
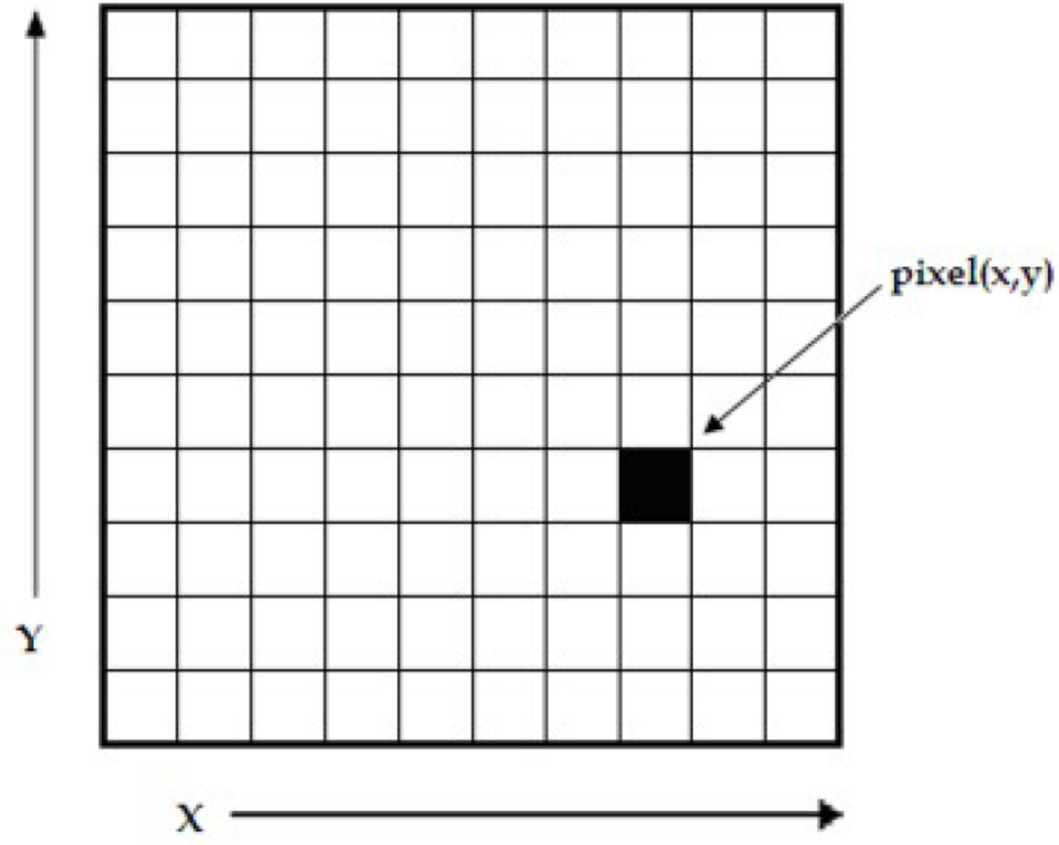


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Image Acquisition Process

- Imaging process is the mapping of 3D object onto a 2D plane
- The object is illuminated by the light from an image sensor.
- The imaging system collect the incoming energy and focus it onto an image plane.
- The intensity of image f at a point (x,y) is proportional to the reflectivity of object and illumination at that point.
- The analog image is converted into digital image by using sampling and quantisation



Representing Digital Images

- The representation of an M×N numerical array as

$$f(x,y)=\begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \dots & \dots & \dots & \dots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

- Each element in the matrix is the pixel value

Basic relationship between pixels

- Neighborhood
- Adjacency
- Connectivity
- Paths
- Regions and boundaries

Neighbors of a Pixel

- N_4 - 4-neighbors (*2 Horizontal, 2 Vertical*)
- N_D - diagonal neighbors
- N_8 - 8-neighbors (N_4 U N_D)

Some basic relationship between pixels

Neighbors of a Pixel

Let p be a pixel at point (x,y)

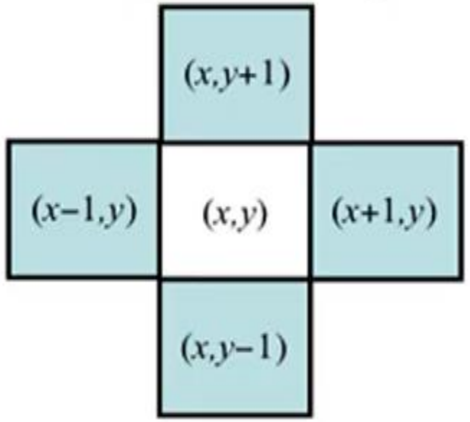
1. **N4 (p) : 4-neighbors of p.**

✓ Any pixel p(x, y) has two vertical and two horizontal neighbors, given by

(x+1,y), (x-1, y), (x, y+1), (x, y-1)

✓ This set of pixels are called the 4-neighbors of P, and is denoted by N4 (P)

✓ Each of them is at a unit distance from P.



4-neighbourhood

- **$N_4(P)$: 4-Neighbors of Pixel P are**
  - **2 vertical neighbors**
  - **2 horizontal neighbors**

$(x-1,y+1)$	$(x,y+1)$	$(x+1,y+1)$
$(x-1,y)$	(x,y)	$(x+1,y)$
$(x-1,y-1)$	$(x,y-1)$	$(x+1,y-1)$

8-neighbourhood

2. **ND(p): diagonal Neighbors**

✓ This set of pixels, called 4 diagonal-neighbors and denoted by ND (p).

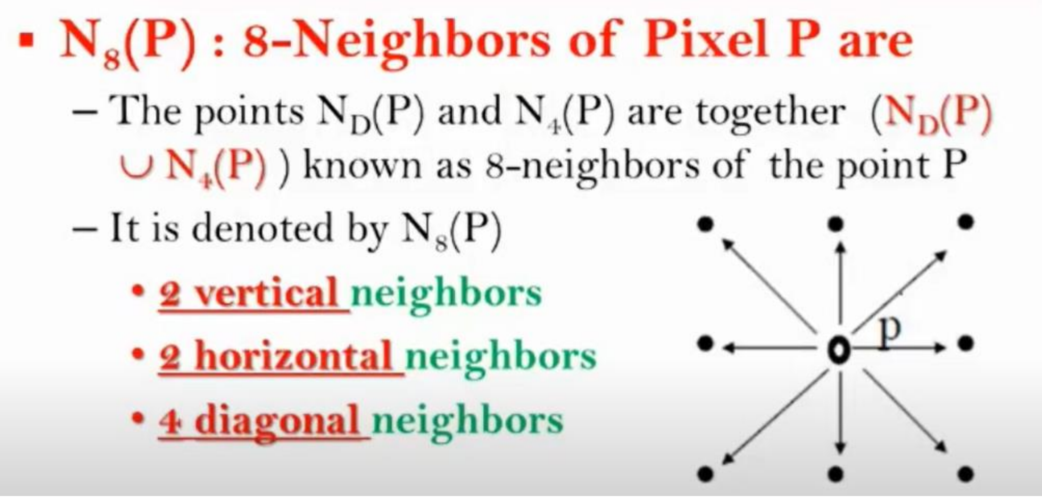
(x+1,y+1), (x+1,y-1), (x-1,y+1), (x-1,y-1)

3. **N8 (p): 8-neighbors of p.**

✓ N4 (P)and ND(p) together are called 8-neighbors of p, denoted by N8 (p).

✓ N8 = N4 U ND

Some of the points in the N4 , ND and N8 may fall outside image when P lies on the border of image.



Neighbors of a Pixel

- N_4 - 4-neighbors (***2 Horizontal, 2 Vertical***)
- N_D - diagonal neighbors
- N_8 - 8-neighbors (N_4 U N_D)

N_D	N_4	N_D
N_4	P	N_4
N_D	N_4	N_D

ADJACENCY

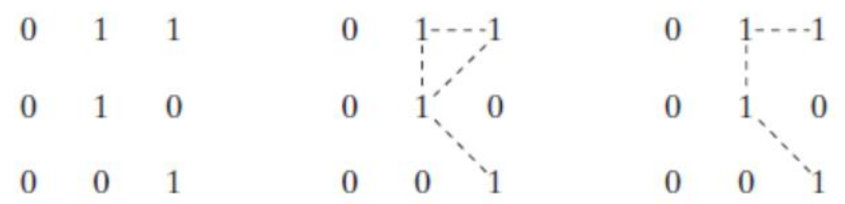
- Two pixels are **connected** if they are neighbors and their gray levels satisfy some specified criterion of similarity.
- For example, in a binary image two pixels are connected if they are 4-neighbors and have same value (0/1)
- **Let v:** a set of intensity values used to ***define adjacency*** and ***connectivity***.
- In a **binary Image v={1},** if we are referring to adjacency of pixels with value 1.
- In a **Gray scale image**, the idea is the same, but **v** typically contains more elements,
- for example **v= {180, 181, 182,....,200}.**
- If the possible intensity values 0 to 255, **v** set could be any subset of these 256 values.

- To understand adjacency let us define V
  - Let V be set of gray levels used to define adjacency

V = {set of gray levels }

Types of Adjacency

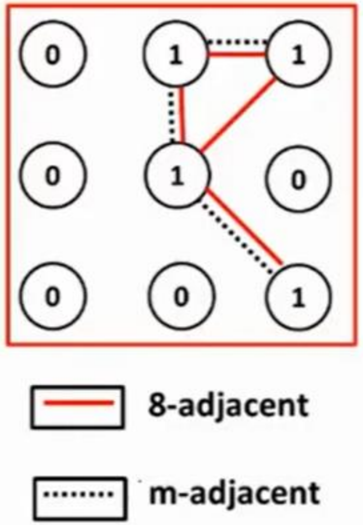
- ✓ **4-adjacency:** Two pixels p and q with values from V are 4-adjacent if q is in the set N4 (p).
- ✓ **8-adjacency:** Two pixels p and q with values from V are 8-adjacent if q is in the set N8 (p).
- ✓ **m-adjacency (mixed):** two pixels p and q with values from V are m-adjacent if:
  - q is in N4 (p) or
  - q is in ND (P) and
  - The set N4 (p) ∩ N4 (q) has no pixel whose values are from v (No intersection).



a b c

FIGURE 2.26 (a) Arrangement of pixels; (b) pixels that are 8-adjacent (shown dashed) to the center pixel; (c) *m*-adjacency.

Adjacency / Connectivity

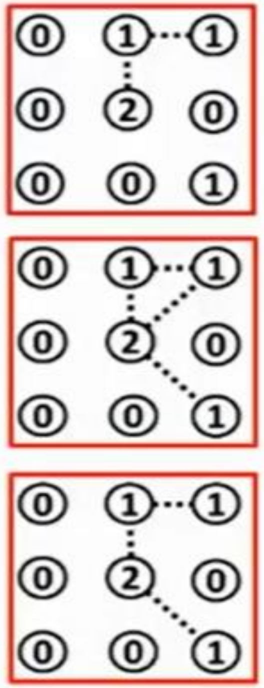


Connectivity

- V be the set of gray-level values used to define connectivity
- Two pixels *p*, *q* that have values from the set V are
  - **4-connected**, if *q* is in the set $N_4(p)$
  - **8-connected**, if *q* is in the set $N_8(p)$
  - **m-connected**, *iff*
    - *q* is in $N4(p)$ or
    - *q* is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ is empty

$V = \{1, 2\}$

(*a*) (*b*) (*c*)



Path

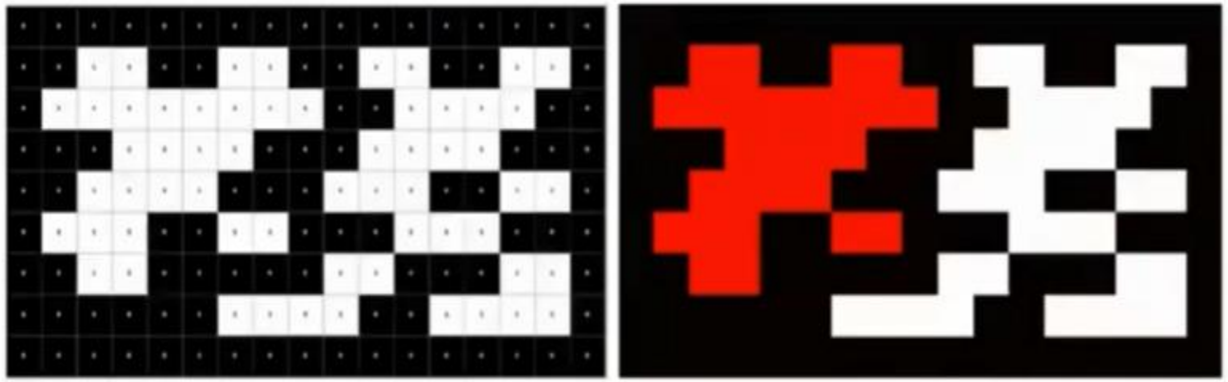
- ✓ A digital path from pixel p(x,y) to pixel q(s,t) is a sequence of distinct pixels with coordinates (x_0 , y_0), (x_1 , y_1), ..., (x_n , y_n), where (x_0 , y_0)= (x,y), (x_n , y_n)= (s,t)
- ✓ (x_i , y_i) is adjacent pixel (x_{i-1}, y_{i-1}) for $1 \le j \le n$,
- ✓ n- The length of the path.
- ✓ If (x_0 , y_0)= (x_n , y_n):the path is closed path.
- ✓ We can define 4,8,m path depending on type of adjacency used

Connected Components

- Connected pixels:
  - If p and q are pixels of an image subset S then p is connected to q in S if there is a path from p to q consisting entirely of pixels in S
- Connected Component
  - For every pixel p in S, the set of pixels in S that are connected to p is called a connected component of S
- Connected Set
  - If S has only one connected component then S is called Connected Set

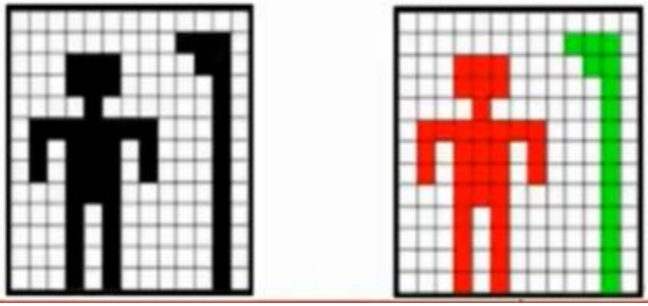
Connected Components - *Examples*

- Connected-component labeling



Input

Output: labeled image



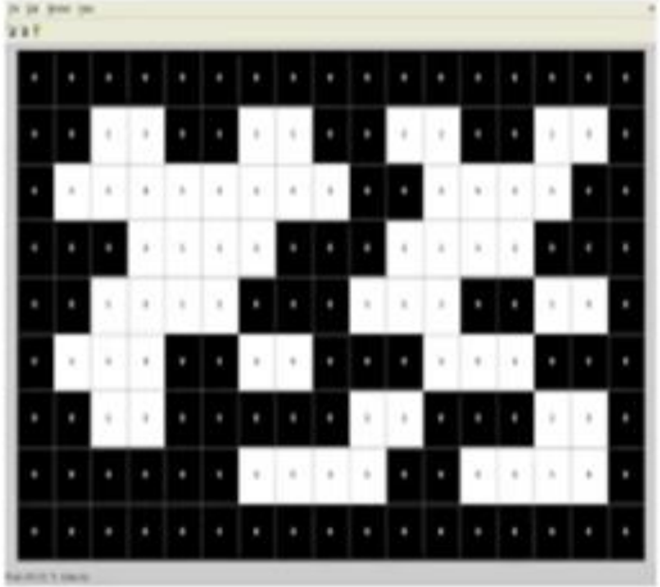
Input

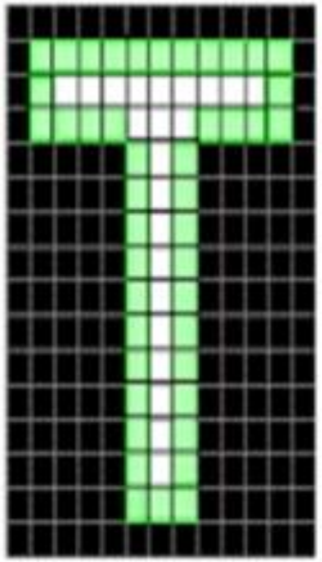
Output:
labeled image

Region and Boundaries

- Region of the image
  - A subset R of pixels in an image is called a *Region of the image if R is a* connected set
    - *Each connected component forms a region*
- Boundary of the Region *R*
  - It is the set of pixels in the region that have one or more neighbors that are not in R
- What if R happens to be entire Image?

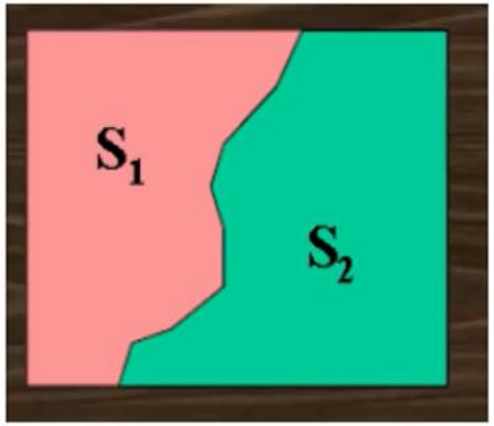
- Two regions, and are said to be adjacent if their 3 union forms a connected set. Regions that are not adjacent are said to be disjoint.
- Suppose that an image contains K disjoint regions, $R_k, k = 1,2,\ldots,K,$ none of which touches the image border. Let R_u denote the union of all the K regions, and let $(R_u)^c$ denote its complement. We call all the points in R_u the foreground And all the points in $(R_u)^c$ the background of the image





CONNECTIVITY

- Let S represent a subset of pixels in an image, Two pixels p and q are said to be connected in S if there exists a path between them.
- Two image subsets S1 and S2 are adjacent if some pixel in S1 is adjacent to some pixel in S2



Consider an image segment shown below.

	3	1	2	1	(q)
	2	2	0	2	
	1	2	1	1	
(p)	1	0	1	2	

(a) Let V={0,1} and compute the length of the shortest 4-,8- and m- path between p and q. If a particular path does not exist between these two points , explain why?

(b) Repeat for V={1,2}.

Sampling and Quantization

- Computer processing of images requires that images be available in digital form. But the output of most of the image sensors is an analog signal
- The analog image is converted into digital image for processing.
- The conversion is achieved in two steps
  - ✓ Sampling
  - ✓ Quantization.

- Continuous image f(x,y) is normally approximated by equally spaced samples arranged in the form of a MxN array .

$$f(x,y)=\begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \dots & \dots & \dots & \dots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

Sampling and Quantization

Sampling

- The process of digitizing the co-ordinate values is called Sampling.
- A sample is the intensity value of an image at a point of time and the samples are taken at each regular intervals.
- The sampling rate is the number of samples taken per second.
- The sampling rate of digitizer determines the spatial resolution of digitized image.

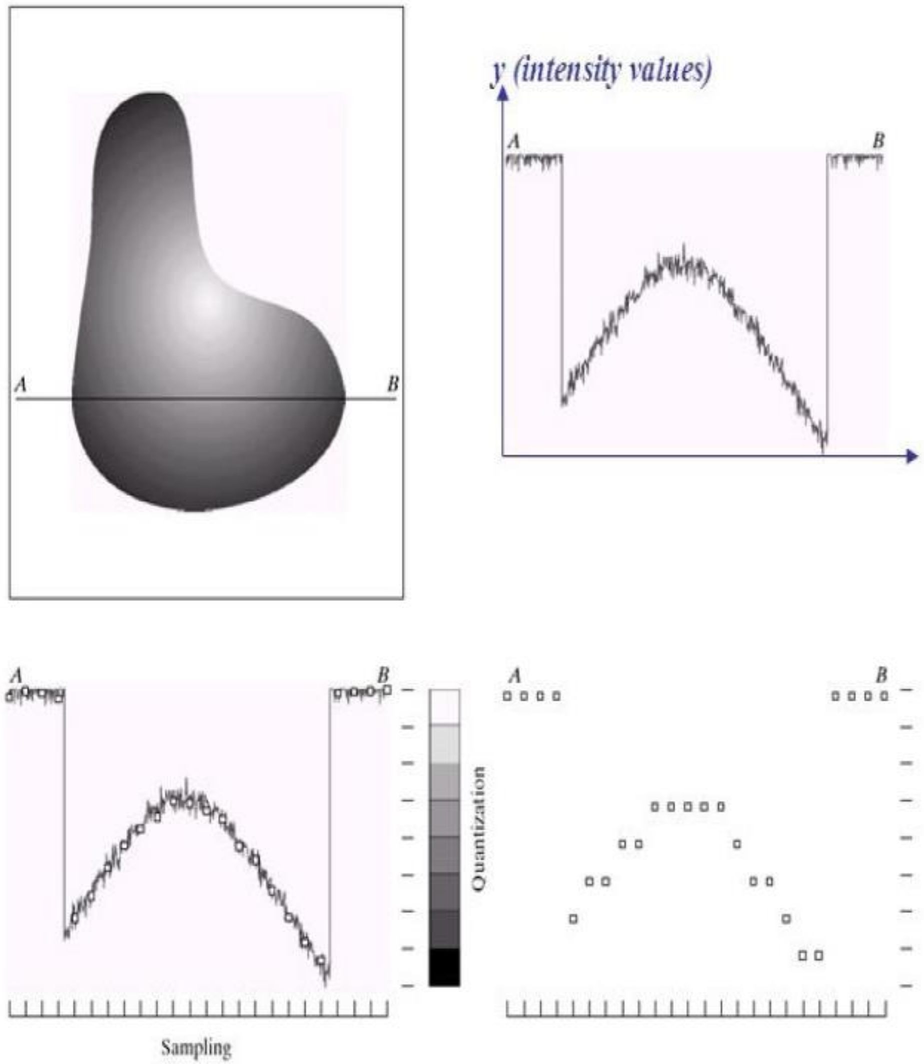
Sampling and Quantization

Quantization

- The process of digitizing the intensity values is called Quantization.
- Approximation of amplitude values to the nearest level is done in quantisation
- Quantization determines the number of grey levels in the digitized images.
- If b-bits are used are used to represent an amplitude,

  No. of quantization levels , $k = 2^b$

  8 bits/pixels are used commonly, that is 256 levels with colors ranging from 0(black) to 255(white).



a b
c d

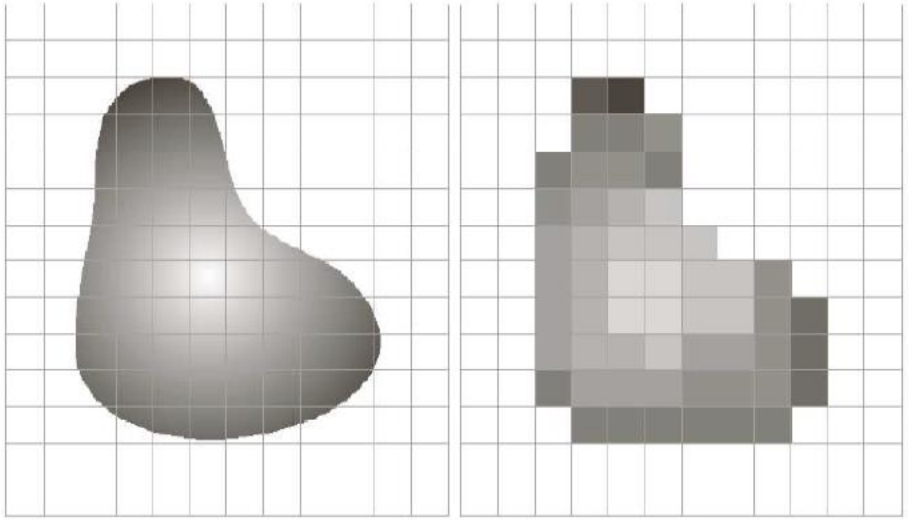
Generating a digital image. (a) Continuous image. (b) A scaling line from A to B in the continuous image, used to illustrate the concepts of sampling and quantization. (c) sampling and quantization. (d) Digital scan line.

Sampling and Quantization

- The basic idea of sampling and quantisation is illustrated in the above figure.
- Fig (a) shows a continues image f(x,y), that we want to convert it into digital form.
- The one-dimensional function in fig(b) is a plot of intensity values of continues image along with a line segment AB in fig (a).
- For sampling, we take equally spaced samples along line AB. The samples are shown as white squares in fig (c).
- The intensity values in fig (c) is still continuous, and we have to apply quantisation on the sampled image

- The figure on the right of (c) shows the intensity level is divided into 8-levels, ranging from black to white.
- The continuous intensity levels are quantised by assigning one of the eight values to each samples as shown in fig(d).
- The 2D digital image can be obtained by carrying out the above procedure line by line starting from top of image.
- The quality of digital image is determined by the number of samples and discrete intensity levels used in sampling and quantisation.

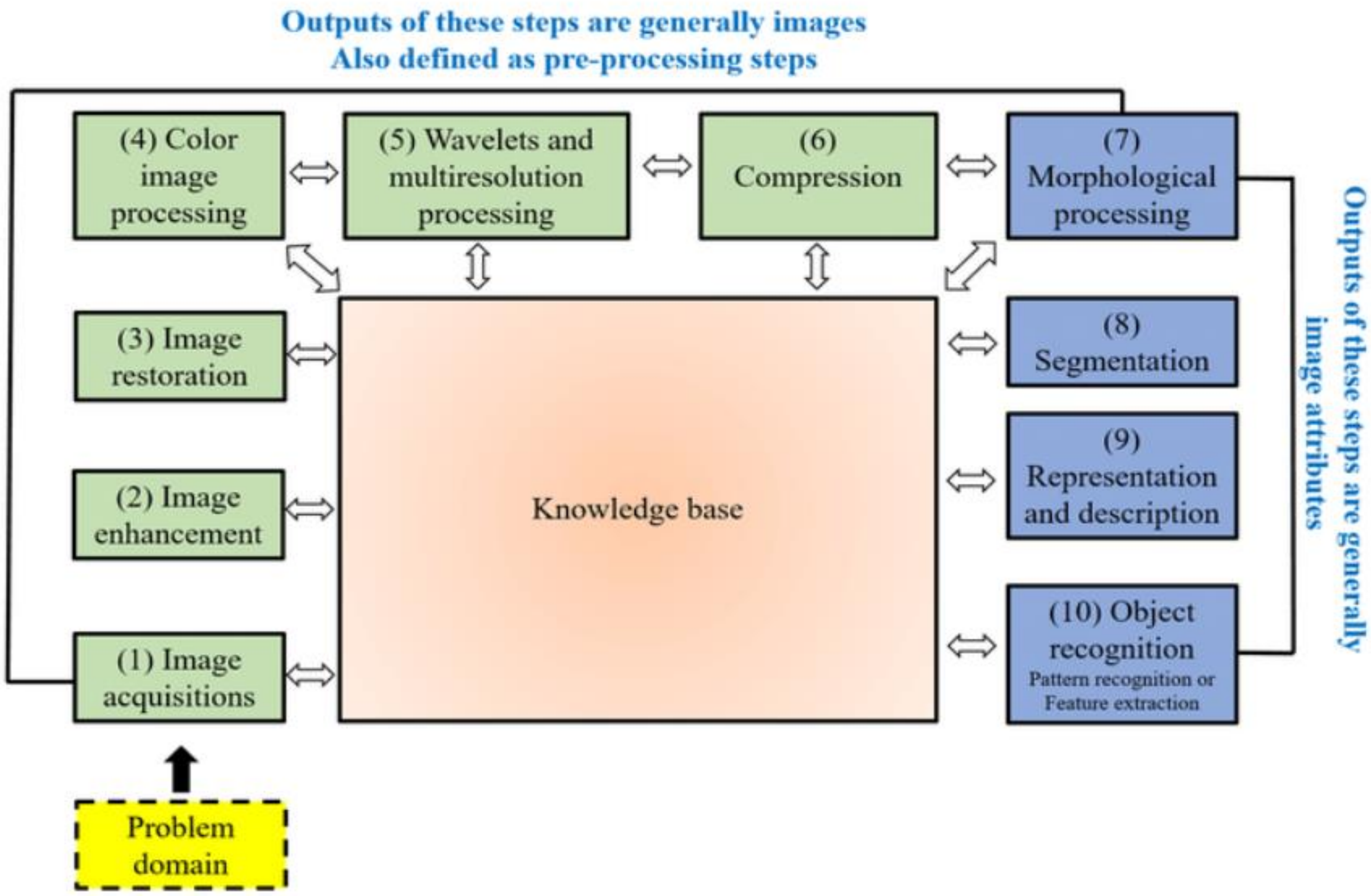
Image Sampling and Quantization



a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

Fundamental steps in digital image processing



Step 1: Image Acquisition

The image is captured by a sensor (eg. Camera), and digitized if the output of the camera or sensor is not already in digital form, using analogue-to-digital convertor

Step 2: Image Enhancement

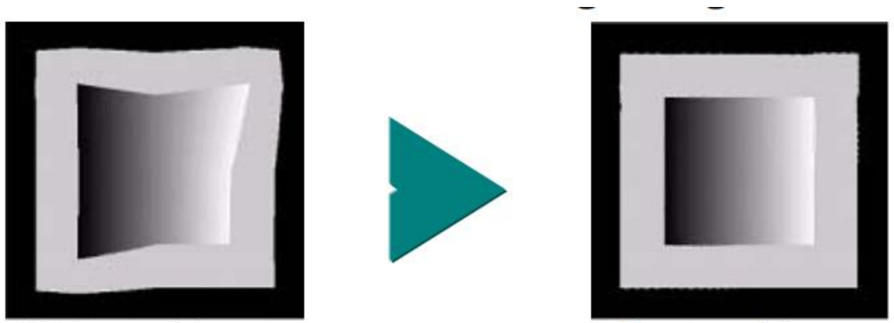
The process of manipulating an image so that the result is more suitable than the original for specific applications.

The idea behind enhancement techniques is to bring out details that are hidden, or simple to highlight certain features of interest in an image.

Step 3: Image Restoration

- Improving the appearance of an image

- Tend to be mathematical or probabilistic models. Enhancement, on the other hand, is based on human subjective preferences regarding what constitutes a "good" enhancement result.



Distorted image Restored image

Step 4: Colour Image Processing

Use the colour of the image to extract features of interest in an image

Step 5: Wavelets

Are the foundation of representing images in various degrees of resolution. It is used for image data compression.

- Wavelets are small waves of limited duration which are used to calculate wavelet transform which provides time-frequency information.
- Wavelets lead to multiresolution processing in which images are represented in various degrees of resolution.

Step 6: Compression

Techniques for reducing the storage required to save an image or the bandwidth required to transmit it.

Step 7: Morphological Processing

Tools for extracting image components that are useful in the representation and description of shape.

In this step, there would be a transition from processes that output images, to processes that output image attributes.

Step 8: Image Segmentation

Segmentation procedures partition an image into its constituent parts or objects.

Step 9: Representation and Description

- **Representation:** Make a decision whether the data should be represented as a boundary or as a complete region. It is almost always follows the output of a segmentation stage.

Boundary Representation: Focus on external shape characteristics, such as corners

Region Representation: Focus on internal properties, such as texture

- **Description:** also called, ***feature selection***, deals with extracting attributes that result in some information of interest.

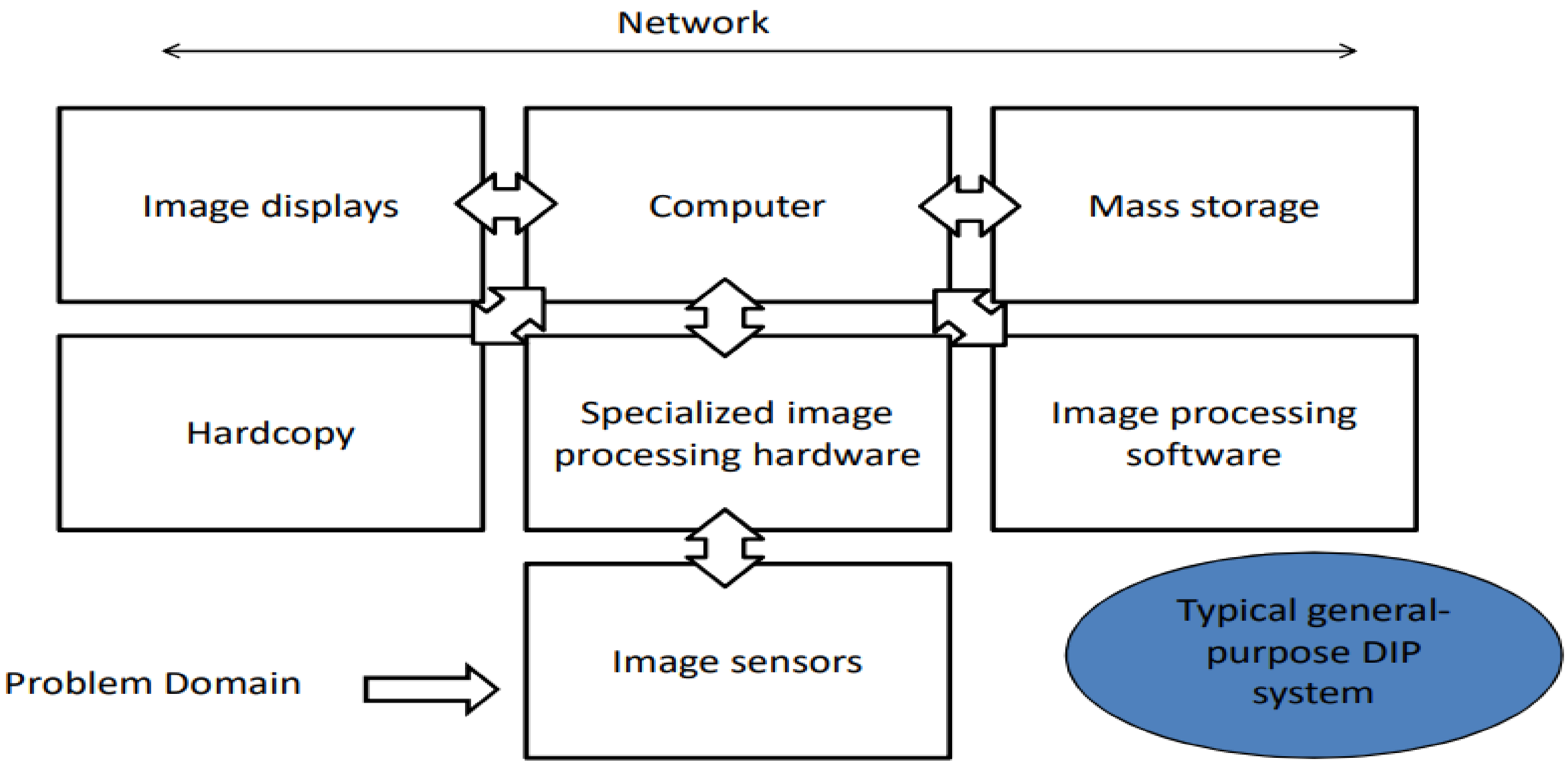
Step 9: Recognition and Interpretation

Recognition: the process that assigns label to an object based on the information provided by its description.

Step 10: Knowledge Base

Knowledge about a problem domain is coded into an image processing system in the form of a knowledge database.

Components of an Image Processing System



1. Image Sensors

Two elements are required to acquire digital images. The first is the physical device that is sensitive to the energy radiated by the object we wish to image (*Sensor*). The second, called a *digitizer*, is a device for converting the output of the physical sensing device into digital form.

2. Specialized Image Processing Hardware

Usually consists of the digitizer, mentioned before, plus hardware that performs other primitive operations, such as an arithmetic logic unit (ALU), which performs arithmetic and logical operations in parallel on entire images.

This type of hardware sometimes is called a front-end subsystem, and its most distinguishing characteristic is speed. In other words, this unit performs functions that require fast data throughputs that the typical main computer cannot handle.

3. Computer

The computer in an image processing system is a general-purpose computer and can range from a PC to a supercomputer. In dedicated applications, sometimes specially designed computers are used to achieve a required level of performance.

4. Image Processing Software

Software for image processing consists of specialized modules that perform specific tasks. A well-designed package also includes the capability for the user to write code that, as a minimum, utilizes the specialized modules.

5. Mass Storage Capability

Mass storage capability is a must in a image processing applications. And image of sized 1024 * 1024 pixels requires one megabyte of storage space if the image is not compressed.

Digital storage for image processing applications falls into three principal categories:

1. Short-term storage for use during processing.
2. on line storage for relatively fast recall
3. Archival storage, characterized by infrequent access

6. Image Displays

The displays in use today are mainly color (preferably flat screen) TV monitors. Monitors are driven by the outputs of the image and graphics display cards that are an integral part of a computer system.

7. Hardcopy devices

Used for recording images, include laser printers, film cameras, heat-sensitive devices, inkjet units and digital units, such as optical and CD-Rom disks.

Networking

- It is a required component to transmit image information over a networked computer.
- Because of the large amount of data inherent in image processing applications, the key consideration in image transmission is bandwidth.